The log-insertion ODE: $g_k'' - \frac{\beta a}{2} g_k' - \nu\beta g_k = k\beta\, g_{k-1}$

Claude, with Daniel Murfet

2026-09-06

Abstract

Unit 15 of the grammar-paper §4 autoformalisation, and the hardest of the log-insertion family: `prop:LogODE`. Writing $g_k(a) = \partial_\lambda^k S_\lambda(a) = \int_0^\infty (\log t)^k t^{\nu-1} e^{-\beta t + \beta a\sqrt{t}}\, dt$, the Weber ODE differentiated k times in the order gives $g_k''(a) - \frac{\beta a}{2} g_k'(a) - \nu\beta g_k(a) = k\beta\, g_{k-1}(a)$. Proved directly for the integral, from an a-derivative ladder and a log-weighted integration-by-parts recurrence. Zero `sorry`/`axiom`.

1 Setting

The general-k order-derivative (unit 13) identified $\partial_\lambda^k S$ with the log-weighted integral g_k = `gLog`. This proposition is the ODE that family satisfies — the order-analogue of the plain Weber ODE (property iii), with an inhomogeneous $k\beta g_{k-1}$ term. It is the last log-insertion item that is achievable without a real-analyticity theory, and the most involved single proof of the whole §4 arc.

2 The things formalised

```
theorem hasDerivAt_gLog (    :  ) (k :  ) (h  : 0 <  ) (h  : 0 <  ) (a :  ) :
    HasDerivAt (fun a => gLog   a   k) (  * gLog   a (  + 1 / 2) k) a

theorem gLog_recurrence (  a   :  ) (k :  ) (h  : 0 <  ) (h  : 0 <  ) :
      * gLog   a (  + 1) k = (  * a / 2) * gLog   a (  + 1 / 2) k
      +   * gLog   a   k + (k :  ) * gLog   a   (k - 1)

theorem gLog_ode (  a   :  ) (k :  ) (h  : 0 <  ) (h  : 0 <  ) :
    deriv (deriv (fun a => gLog   a   k)) a - (  * a / 2) * deriv (fun a => gLog   a   k) a
      -   *   * gLog   a   k = (k :  ) *   * gLog   a   (k - 1)
```

3 Proof strategy

Two ingredients, assembled by the plain-Weber logic.

- **a-derivative ladder.** $\partial_a e^{-\beta t+\beta a\sqrt{t}} = \beta\sqrt{t}\, e^{\cdots}$, so differentiating `gLog` under the integral in a multiplies the integrand by $\beta\sqrt{t} = \beta t^{1/2}$, i.e. shifts the order by $\frac{1}{2}$: ∂_a`gLog`$(\nu, k) = \beta\,$`gLog`$(\nu + \frac{1}{2}, k)$. This is `hasDerivAt_fluctuation` with the $(\log t)^k$ weight carried along; two applications give $g_k'' = \beta^2$`gLog`$(\nu + 1, k)$.

- **Log-weighted recurrence.** From $\int_0^\infty \partial_t[t^\nu(\log t)^k e^{\cdots}]\,dt = 0$ (improper FTC). The t-derivative has *four* terms: the usual ν, $-\beta$, $\frac{\beta a}{2}$ pieces plus $k\,t^{\nu-1}(\log t)^{k-1}e^{\cdots}$ from differentiating the log weight. Matching the four to gLog integrands gives $\beta\,\texttt{gLog}(\nu+1,k) = \frac{\beta a}{2}\texttt{gLog}(\nu+\frac{1}{2},k) + \nu\,\texttt{gLog}(\nu,k) + k\,\texttt{gLog}(\nu,k-1)$.

Substituting the ladder into the ODE reduces it, after dividing by β, to exactly the recurrence.

4 Roadblocks and resolutions

- **Boundary limits with a logarithm.** The improper FTC needs $t^\nu(\log t)^k e^{\cdots} \to 0$ at both 0^+ and ∞, but $(\log t)^k$ is unbounded at each end. The clean squeeze is $|t^\nu(\log t)^k| \le (k/\delta)^k(t^{\nu+\delta}+t^{\nu-\delta})$ (`abs_log_pow_le`) with $\delta=\nu/2$, giving exponents $\frac{3\nu}{2}, \frac{\nu}{2}$ — both *strictly positive*, so the majorant $\to 0$ at 0^+ (continuity, value 0) and at ∞ (Gaussian decay, `tendsto_rpow_mul_exp_neg_mul_atTop_nhds_zero`). Note $\delta = c/2$ (fine for *integrability*) fails here for $\nu<1$: the boundary needs the exponent >0, not just >-1.
- The four-term integral decomposition triggers the `Pi.add` pattern-match gotcha under `integral_add`; fixed with type-ascribed single-lambda `IntegrableOn` witnesses.
- Small `rpow` identities ($t^\nu t^{-1} = t^{\nu-1}$, $t^\nu(2\sqrt{t})^{-1} = \frac{1}{2}t^{\nu-1/2}$) and the final constant algebra ($(t^{\nu/2}+t^{-\nu/2})t^\nu = t^{3\nu/2}+t^{\nu/2}$) go through `Real.rpow_add` plus a `linear_combination` over the two exponent identities.

5 What was Mathlib, what was new

Mathlib supplied the parametric-FTC and improper-FTC lemmas, `Real.hasDerivAt_log`, `HasDerivAt.pow`, the rpow/sqrt derivatives, and `tendsto_rpow_mul_exp_neg_mul`. New: the a-derivative ladder for the log-weighted integral, the log-weighted IBP recurrence with its extra kg_{k-1} term, the two-sided log-power boundary squeeze, and the assembled ODE — the order-analogue of the Weber equation.

6 Lessons

The log weight changes two things versus the plain recurrence: it adds one term to the pointwise t-derivative (giving the inhomogeneous kg_{k-1}), and it makes the boundary limits nontrivial. The key is to bound the *whole* boundary integrand $t^\nu|\log t|^k$ (not the $t^{\nu-1}$ integrand) by a symmetric power sandwich with $\delta<\nu$, so both exponents stay positive and vanish at 0^+ — the same lemma used at $\delta=c/2$ for integrability serves at $\delta=\nu/2$ for the limits.

7 Follow-ups

Of the remaining §4 items, `lem:log_generating` ($\sum \frac{z^p}{p!}(c-\partial_\nu)^p S_\nu = e^{zc}S_{\nu-z}$) is Taylor's theorem in the order and needs real-analyticity of S_ν in ν; the ladder-operator commutator $[b,b^\dagger]=1$ and number operator want a Weyl-algebra framework.